

PROJECT ABSTRACT

Smart Waste Reporting & Management System

1. Problem Statement (Existing System)

Public waste issues such as roadside garbage, overflowing waste bins, and illegal dumping are common in many municipalities. Existing complaint systems mainly depend on phone calls, office visits, or social media, making the process slow and inefficient. Citizens cannot easily track the status of their complaints, while municipality officials face challenges in assigning sanitation workers, monitoring cleaning activities, managing staff, and ensuring timely waste clearance. The lack of a centralized digital platform results in delayed responses, poor coordination, and reduced accountability.

2. Objective

The objective of this project is to develop a web-based waste reporting and management system that enables citizens to report public waste complaints online, helps municipalities manage complaints efficiently, assign sanitation workers or worker teams, monitor staff performance, and ensure waste complaints are resolved within 1–2 days. The system also aims to improve transparency through real-time complaint tracking, notifications, and effective staff management.

3. Proposed Solution

The proposed application provides three user roles: **Citizen, Admin (Municipality), and Sanitation Worker.**

Citizens can register, log in securely, and report waste complaints by uploading a photo, entering a description, and selecting the waste location on an interactive map. They can track the complaint status, receive email and in-app notifications, and view their complaint history.

The **Municipality Admin** verifies complaints, creates multiple cleaning teams, adds multiple sanitation workers to each team, and generates unique usernames and passwords for every worker. The admin assigns complaints to an individual worker or a worker team based on the size of the complaint and worker availability. Each complaint is given a mandatory 1–2 day completion deadline. The admin monitors complaint progress, manages staff performance, tracks worker availability (Online/Offline), reassigns overdue complaints when necessary, and generates reports.

Sanitation workers log in using the credentials provided by the municipality. They can switch their availability between Online and Offline, view assigned complaints, update work status (Assigned → In Progress → Completed), upload after-cleaning photos as proof of completion, and view their completed task history.

After the municipality verifies the completed work, the system automatically sends an email notification and an in-app notification to the citizen confirming that the complaint has been resolved. The municipality can also provide a performance bonus for each successfully completed task. This bonus is recorded in the system and can be viewed by the respective worker in their dashboard.

4. Technology Stack

Frontend: React.js

Backend: Node.js, Express.js

Database: MongoDB

Authentication: JWT

Maps: Leaflet (OpenStreetMap)

Image Storage: Cloudinary

Email Notifications: Nodemailer

5. Benefits of the Project

Enables citizens to report waste complaints online.

Reduces manual complaint handling and improves response time.

Ensures complaints are resolved within 1–2 days.

Provides real-time complaint tracking and notifications.

Improves municipality staff management and work allocation.

Supports both individual worker and worker team assignments.

Tracks worker availability and completed tasks.

Sends automatic email and in-app notifications after complaint resolution.

Records performance-based bonus payments (dummy) for completed tasks.

Provides photo-based proof of completed cleaning for better transparency and accountability.

Helps municipalities maintain cleaner public spaces through an efficient digital waste management system.

MODULE DESCRIPTION

Citizen Module

- Registration
- Secure Login
- Dashboard
- Report Waste Complaint (Photo + Description + Map Location)
- Track Complaint Status
- View Complaint History
- Receive Email Notifications
- Receive In-App Notifications
- Update Profile

Sanitation Worker Module

- Secure Login (Admin-provided credentials)
- Switch Online / Offline
- View Assigned Team
- View Assigned Complaints
- Accept Assigned Tasks
- Update Complaint Status
- Upload After-Cleaning Photos
- View Completed Task History
- View Performance Bonus History (Dummy)
- Receive Notifications
- Update Profile

Admin (Municipality) Module

- Secure Login
- Dashboard
- Verify / Reject Complaints
- Create Cleaning Teams
- Add Multiple Workers to Teams
- Generate Username & Password for Workers
- Edit / Delete Worker Accounts
- Assign Complaints to Workers/Teams
- Monitor Worker Availability (Online/Offline)
- Set 1–2 Day Completion Deadline
- Track Complaint Progress
- Reassign Overdue Complaints
- Verify Completed Work
- Send Email & In-App Notifications to Citizens
- Provide Performance Bonus (Dummy Payment) for Completed Tasks
- View Worker Performance
- Generate Reports
- Manage Announcements